

Observation of Dual-Band Intrinsic Chirality in Underetched Silicon Metasurfaces via Quasi-BICs

Leyi Zhu, Yukang Zhang, Guanhua Yang, Rong Jin, Chenyang Wang, Yueke Wang, Guanhai Li,* and Tian Sang*



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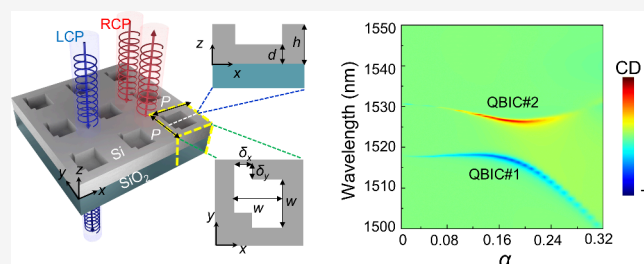
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ABSTRACT: Intrinsic chirality via bound states in the continuum (BICs) offers a promising approach to spin-selective light–matter interactions but often relies on complex three-dimensional (3D) or 2.5D geometries that hinder CMOS compatibility. Here we demonstrate a fabrication-friendly strategy for intrinsic chiral quasi-BICs (QBICs) in silicon (Si) metasurfaces by exploiting a single-step underetch process to introduce controllable vertical asymmetry. Combined with mild in-plane perturbation, this enables independent control over two symmetry-breaking pathways, yielding dual-band intrinsic circular dichroism (CD) with a tunable sign and magnitude. We develop a chiral coupled-mode theory (CCMT) for universal design rules linking the sign of CD to the spinning-dependent coupling phase for maximizing intrinsic chirality. Guided by this framework, we experimentally demonstrate near-unity intrinsic chirality ($CD = 0.90$) and an ultrahigh quality factor exceeding 3000 in the telecom band. Our findings provide a CMOS-compatible route to robust intrinsic chirality and offer a versatile platform for advanced chiral metadevices.

KEYWORDS: *intrinsic chirality, bound states in the continuum, circular dichroism, underetched silicon metasurfaces*



Optical chirality—the ability of a nanostructure to discriminate between left- and right-handed circularly polarized light—lies at the heart of diverse nanophotonic applications.^{1–3} It is quantified by the standard metric of circular dichroism (CD), which measures the differential interaction with electromagnetic fields of opposite handedness.^{4–6} Unfortunately, the inherently weak chiroptical response of natural materials and nanostructures results in limited CD signals due to the mismatch between their size and the exciting wavelength.^{7,8} Although strong CD can be achieved in planar structures by breaking in-plane symmetry—via structural anisotropy or oblique illumination—this approach only induces *extrinsic* (false) chirality.^{9–12} Such chirality is an artifact of the measurement geometry and not an intrinsic property, thus hindering its utility in practical applications such as chiral light emission and polarization photodetection.

To achieve significant *intrinsic* (true) chirality, three-dimensional (3D) metamaterials such as helical structures,^{13–15} 3D-L particles,¹⁶ and spherical crowns¹⁷ are proposed to enhance the chiroptical responses. The recent emergence of chiral quasi-bound states in the continuum (QBIC), induced by chiral perturbations, has enabled the engineering of geometric phases and the encoding of polarization states in planar metasurfaces.^{18,19} By converting a BIC into a quasi-BIC (QBIC) through controlled symmetry breaking, one can achieve ultrahigh-Q resonances whose

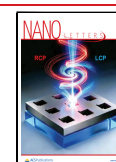
radiation channels couple selectively to a single optical helicity in both dielectric²⁰ and plasmonic²¹ metasurfaces. Such intrinsic chiral QBICs—whose handedness is determined entirely by the structure rather than illumination geometry—have enabled near-unity CD and strong spin–photon interactions on subwavelength scales. Recent theoretical and experimental advances have demonstrated that realizing intrinsic chirality in 2.5D nanostructures requires not only in-plane symmetry breaking but also deliberate control over *out-of-plane* degrees of freedom, which govern the interference of multipolar radiation and the phase structure of BIC leakage channels. Landmark works in this direction include metasurfaces employing slanted dielectric geometries^{22–24} to achieve asymmetric vertical coupling, as well as complex multiheight architectures^{25,26} that “unlock” the out-of-plane dimension to approach single-helicity radiation conditions. Despite these breakthroughs, existing implementations of intrinsic chirality generally depend on complex 2.5D meta-atoms, fabricated through techniques like slant-etching or multistep lithography and deposition processes. The sample

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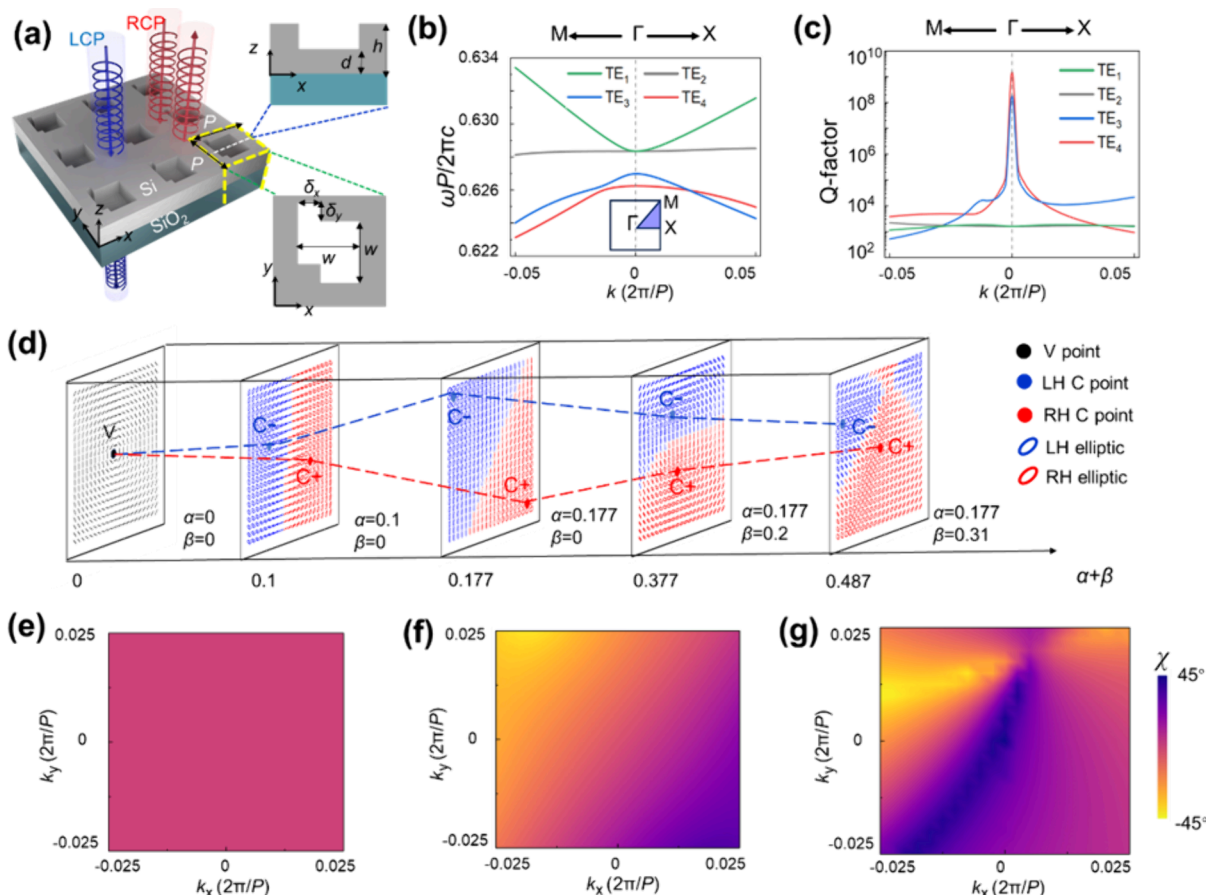


Figure 1. Origin of intrinsic chirality in the USM. (a) Schematic of the underetched metasurface to achieve intrinsic chiral BICs. The x - z and x - y planes of the unit cells are zoomed in the inset of the figure. The geometric parameters are $P = 905$ nm, $h = 500$ nm, and $w = 480$ nm. (b) Band structure and (c) Q-factor of the nonperturbation metasurface with $(\alpha, \beta) = (0, 0)$. (d) Evolution of the C points of the TE_4 band in the k -space for different α and β . Distribution maps of the ellipticity angle χ of the metasurface for (e) $(\alpha, \beta) = (0, 0)$, (f) $(\alpha, \beta) = (0.177, 0)$, and (g) $(\alpha, \beta) = (0.177, 0.31)$.

size in the slant-etching system²³ is constrained by the limited capacity of electron-beam lithography (EBL), which requires dividing larger patterns into separately exposed subareas and inevitably causes alignment dislocation at the boundaries. Additionally, fabricating multiheight architectures demands more specialized techniques such as grayscale EBL²⁵ or additional deposition steps,²⁶ introducing complexities that pose substantial barriers to reproducibility, large-area patterning, and monolithic integration on CMOS-compatible silicon (Si) platforms.

In this work, we introduce an experimentally accessible and CMOS-compatible route to engineer intrinsic chiral QBICs by leveraging a single-step underetch process on a silicon-on-insulator (SOI) metasurface. In contrast to previous chiral meta-atoms, the proposed approach based on 2.5D nanostructures generates an effective out-of-plane asymmetry by selectively removing the top Si layer, thereby tailoring the circular polarization singularities (C-points) to the Γ -point by combination with the in-plane perturbation. We show that such a fabrication-friendly underetched silicon metasurface (USM) naturally supports dual high-Q chiral QBICs exhibiting opposite CD signs. The far-field intrinsic chirality of the USM is characterized by the near-field volume-integrated total reactive helicity density (vt-RHD), with the underlying physical mechanism attributed to spinning-selected magnetic dipole (MD) modes. We further develop a chiral coupled-

mode theory (CCMT) framework to model dual-band CD spectra with opposite signs for the two QBIC modes, achieving excellent agreement between theoretical predictions and simulations. Finally, we experimentally demonstrate near-unity intrinsic chirality ($CD = 0.90$) and ultrahigh quality factors ($Q > 3000$) in the telecom band. Our work establishes a lithography-only, SOI- and CMOS-compatible strategy for realizing robust intrinsic chiral BICs and offers promising opportunities for integrated chiral sensing, spin-selective emitters, nonlinear chiral processes, and large-area chiral photonic devices.

Figure 1 illustrates the principle of the USM for the generation of intrinsic chirality. Figure 1a shows the schematic diagram of the underetched metasurface to achieve intrinsic chiral BICs, where a Si nanoarray with period P is supported by a fused silica (SiO_2) substrate. The refractive indices of Si and SiO_2 in the wavelength region of interest are 3.48 and 1.47, respectively. The unit cell of the initial Si nanoarray is a squared nanohole with width w , and its in-plane mirror symmetry is broken by shifting an edged-rectangle nanohole along the y direction. The side lengths of the edged-rectangle nanohole along the x and y direction are δ_x and δ_y , respectively; and $\delta_y = 0.5\delta_x + 0.125w$. The in-plane asymmetric parameter is defined as $\alpha = \delta_x/w$. The out-of-plane mirror symmetry is broken by the underetching with thickness d , and the out-of-plane asymmetric parameter is defined as $\beta = d/h$. Therefore,

all mirror symmetries of the structure are broken if both α and β are nonzero. Figure 1b shows the band structure of the nonperturbation Si metasurface with $(\alpha, \beta) = (0, 0)$. A series of Bloch modes indicated as the TE₁–TE₄ bands are supported by the metasurface, and these TE-like modes are dominated by the in-plane electric fields with negligible H_x , H_y , and E_z components (see Supporting Information Section 1 and Figure S1). Numerical simulations were performed using commercial software COMSOL Multiphysics to study the eigenmodes. In Figure 1c, it can be seen that the TE₃ and TE₄ bands are associated with two symmetry-protected BIC modes, featuring an infinite Q-factor at the Γ point, and they can be transformed into high-Q QBICs by breaking the symmetry of the structure.^{27–31} To elucidate the topological natures of chiroptical behaviors of the USM in the momentum space (k -space), the far-field polarization vector on the x – y plane is introduced as $\mathbf{d}(\mathbf{k}_{\parallel}) = d_x(\mathbf{k}_{\parallel})\hat{x} + d_y(\mathbf{k}_{\parallel})\hat{y}$, and the topological charge q carried by the polarization singularity (V point) or C point can be obtained as^{32,33}

$$q = \frac{1}{2\pi} \oint_L d\mathbf{k}_{\parallel} \cdot \nabla_{\mathbf{k}_{\parallel}} \phi(\mathbf{k}_{\parallel}) \quad (1)$$

where L is a loop along the counterclockwise direction around the BIC in $\mathbf{k}_{\parallel}$ space and $\phi(\mathbf{k}_{\parallel})$ is the angle of the polarization vector with $\phi(\mathbf{k}_{\parallel}) = \frac{1}{2} \arg[S_1(\mathbf{k}_{\parallel}) + iS_2(\mathbf{k}_{\parallel})]$, where $S_i(\mathbf{k})$ is the Stokes parameter of $\mathbf{d}(\mathbf{k}_{\parallel})$.

Figure 1d illustrates the evolution of the C point of the TE₄ band in k -space by tuning both α and β . For the nonperturbation metasurface with $(\alpha, \beta) = (0, 0)$, BIC manifests as a V point with topological charge $q = +1$ at the Γ point. When the in-plane perturbation α is introduced, this V point dissociates into a pair of right-handed (RH) and left-handed (LH) C points with an identical topological charge of $q = +1/2$, thereby conserving the global topological charge. Furthermore, the out-of-plane perturbation β through under-etching provides an additional degree of freedom to control this C-point pair. By simultaneously varying α and β , their evolutionary trajectories in k -space can be precisely tailored. Notably, at $(\alpha, \beta) = (0.177, 0.31)$, the RH C point is tuned directly to the Γ point, signifying the realization of an intrinsic chiral QBIC. Figure 1e–g illustrates the distribution maps of the ellipticity angle χ of the metasurfaces with different ellipticities (α, β) . The ellipticity angle χ is given by $\chi = \arctan(n/m)$, where m and n are the lengths of the semimajor and semiminor axes of the polarization ellipse, respectively.³⁴ The ellipticity angle χ takes values between -45° and $+45^\circ$. $\chi = 0^\circ$ corresponds to the linear polarization state, and $\chi = +45^\circ$ (-45°) corresponds to the RH (LH) circular states, while other χ values represent RH and LH elliptical states. As shown in Figure 1e, the far-field radiation at $(\alpha, \beta) = (0, 0)$ exhibits a linearly polarized state with $\chi = 0^\circ$ in the k -space. Introducing in-plane asymmetry ($\alpha = 0.177$) perturbs this state, producing the RH (LH) circular and elliptical polarizations, as shown in Figure 1f. Finally, as can be seen in Figure 1g, adding the out-of-plane asymmetry ($\beta = 0.31$) yields an RH circular state ($\chi = +45^\circ$) at the Γ point. Note in the case of overetching, the intrinsic chiral QBIC is inaccessible due to small out-of-plane perturbation from the low-index SiO₂ layer (see Supporting Information Section 2 and Figure S2).

To gain a comprehensive understanding of the intrinsic chirality between the near-field distributions and the far-field chiroptical responses of the USM, the RHD^{35,36} is employed to

evaluate the chirality: $\text{RHD} = c/\omega_0 \text{Re}(\mathbf{B}^* \cdot \mathbf{D})$, where c denotes the speed of light in a vacuum, ω_0 represents the angular frequency of the incident light, $\mathbf{B}^*$ is the complex conjugate of the magnetic field, and $\mathbf{D}$ is the electric displacement vector. The RHD serves as the dominant role for the generation of chiral radiation in resonant metasurfaces, and the metasurfaces can support a nonzero RHD when both the in-plane and out-of-plane symmetries are broken.³⁷ Figure 2a shows the

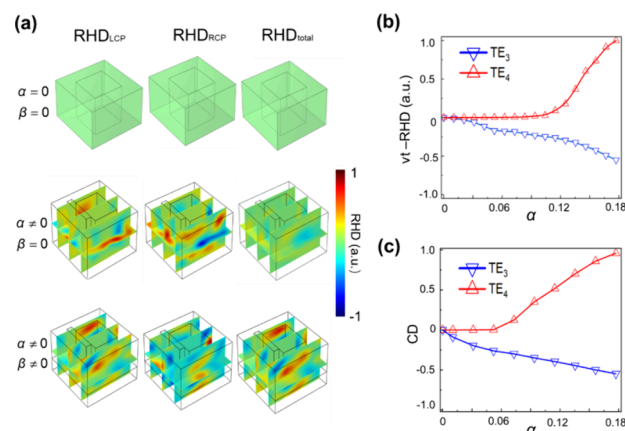


Figure 2. Near-field RHD and far-field properties through symmetry breaking. (a) Distributions of the normalized near-field RHD of the metasurface under illumination of the LCP and RCP light as α and β are varied, where $\text{RHD}_{\text{total}} = \text{RHD}_{\text{LCP}} + \text{RHD}_{\text{RCP}}$. (b) vt-RHD as a function of α with $\beta = 0.31$ for the resonances of the TE₃ and TE₄ bands. (c) CD value as a function of α with $\beta = 0.31$ for the resonances of the TE₃ and TE₄ bands.

distributions of the normalized near-field RHD of the metasurface as α and β are varied; other structural parameters are the same as in Figure 1a. As shown in Figure 2a, the near-field RHD is zero at $(\alpha, \beta) = (0, 0)$ due to the achiral nature of the pristine metasurface. Introducing in-plane perturbation ($\alpha \neq 0$) generates and significantly enhances the RHD for both left-handed circularly polarized (LCP) and right-handed circularly polarized (RCP) excitations. However, the distributions of the RHD for the LCP and RCP lights are nearly opposite, resulting in a weak total RHD due to this cancellation. Consequently, introducing out-of-plane perturbation ($\beta \neq 0$) produces a significant total RHD, as the enhanced RHD for both LCP and RCP excitations exhibits similar distributions. Note that on comparing with optical chiral density (OCD) defined as $\text{OCD} = c/\omega_0 \text{Im}(\mathbf{B}^* \cdot \mathbf{D})$, RHD plays a dominant role due to the high-Q resonance features of QBICs (see Supporting Information Section 3 and Figure S3).

To corroborate the correlation between the RHD distributions and intrinsic chirality of the USM, we introduce the vt-RHD for the volume-integration of both the LCP and RCP excitations as

$$\iiint_V \text{RHD}_{\text{total}} dV = \iiint_V \text{RHD}_{\text{LCP}} dV + \iiint_V \text{RHD}_{\text{RCP}} dV \quad (2)$$

where V is a finite volume that includes the Si metasurface and surrounding background and RHD_{LCP} and RHD_{RCP} are the RHD for the LCP and RCP light, respectively. Similarly, Figure 2b shows the vt-RHD for the resonances of the TE₃ and TE₄ bands as α is varied with $\beta = 0.31$. As can be seen in Figure 2b, the vt-RHD is zero for the metasurface, with only out-of-plane

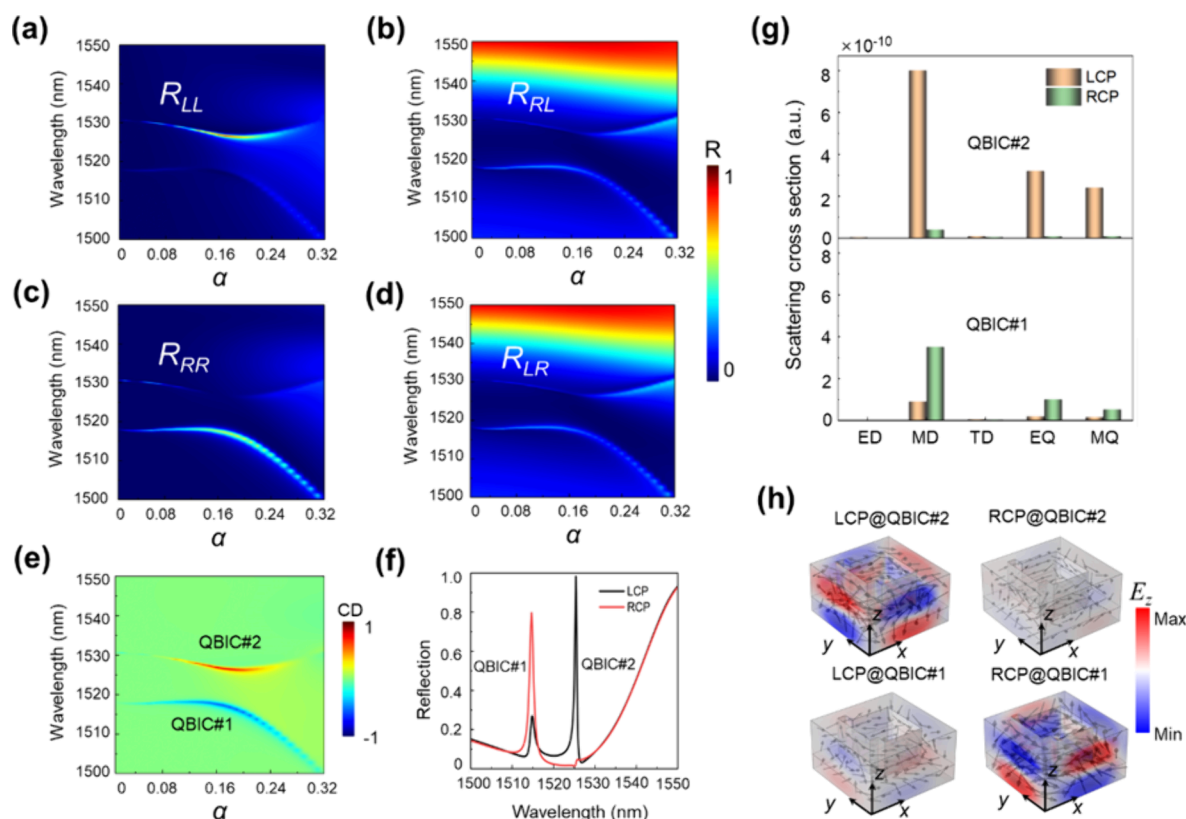


Figure 3. Resonant properties of the high-Q intrinsic CD of the USM. (a) Co-polarized R_{LL} and (b) cross-polarized R_{RL} components of the reflected LCP light as functions of α with $\beta = 0.31$. (c) Co-polarized R_{RR} and (d) cross-polarized R_{LR} components of the reflected RCP light as functions of α with $\beta = 0.31$. (e) CD responses as functions of α with $\beta = 0.31$. (f) Reflection spectra of the LCP and RCP light at $\alpha = 0.177$ with $\beta = 0.31$, revealing two distinct chiral QBICs with QBIC#1 at 1515.7 nm and QBIC#2 at 1526 nm. (g) Multipole decomposition at the resonant locations of QBIC#1 and QBIC#2 at $\alpha = 0.177$ with $\beta = 0.31$. (h) Near-field distributions in the unit cell of the USM at the resonance locations of QBIC#1 and QBIC#2 at $\alpha = 0.177$ with $\beta = 0.31$. Black arrows indicate the displacement current vector.

symmetry breaking at $\alpha = 0$. The introduction of the in-plane symmetry breaking with $\alpha \neq 0$ will result in a nonzero vt-RHD. For both the TE_3 and TE_4 bands, the magnitude of the vt-RHD increases gradually with the increase of α , and the two bands exhibit opposite signs due to their distinct vt-RHD distributions. Particularly, the absolute value of the vt-RHD for the TE_4 resonance is larger than that of the TE_3 resonance, indicating its stronger chiroptical responses. Figure 2c illustrates the CD value as a function of α with $\beta = 0.31$ for the resonances of the TE_3 and TE_4 bands. CD is defined as $CD = R_{LCP} - R_{RCP}$, representing the difference in reflection between LCP and RCP light. As can be seen in Figure 2c, the CD values and their signs for both the TE_3 and TE_4 resonances follow the trend of the vt-RHD distributions, confirming the fundamentally important role of vt-RHD in generating intrinsic chirality in the USM.

To examine the resonant characteristics of the high-Q intrinsic CD in the USM, we analyze the reflections of the LCP and RCP light by decomposing them into copolarized and cross-polarized components. This decomposition is defined as $R_{LCP} = R_{LL} + R_{RL}$ and $R_{RCP} = R_{RR} + R_{LR}$, as shown in Figure 3a–d. It can be seen that the copolarized components differ significantly at the resonance locations of the two QBICs associated with the TE_3 and TE_4 bands. In contrast, the cross-polarized components for both LCP and RCP light are identical, indicating the structure's inherent chirality of the USM.³⁸ Similarly, the transmission of the USM also exhibits high-Q intrinsic chirality with identical cross-polarized

components (see Supporting Information, Section 4 and Figure S4). Additionally, due to the inherent chiral properties of the USM, reciprocal excitation along the $+z$ versus $-z$ directions for both reflection and transmission can be achieved (see Supporting Information Section 5 and Figure S5), which is distinctly different from that of the extrinsic chirality.³⁹ Figure 3e presents the CD responses as functions of α with β fixed at 0.31, showing the emergence of two high-Q intrinsic CDs with opposite signs at the resonance locations of the two QBICs as α increases. Figure 3f displays the reflection spectra of the USM for the LCP and RCP light at $\alpha = 0.177$ with $\beta = 0.31$, revealing two distinct chiral QBICs with QBIC#1 at 1515.7 nm and QBIC#2 at 1526 nm.

We further analyze the far-field scattered power to elucidate the resonance mechanisms of the USM according to the multipole decomposition theory.^{40–42} This includes the contributions of the scattered power from the electric dipole (ED), magnetic dipole (MD), toroidal dipole (TD), electric quadrupole (EQ), and magnetic quadrupole (MQ) (see Supporting Information Section 6). As shown in Figure 3g, the magnetic dipole (MD) is the dominant mode for both the LCP and RCP excitations at the resonance locations of QBIC#1 and QBIC#2. Notably, the scattered power of the MD mode for the RCP light significantly exceeds that for the LCP light at QBIC#1; by contrast, this asymmetry completely reverses at QBIC#2. Figure 3h illustrates the near-field distributions in the unit cell of the USM at the resonance locations of QBIC#1 and QBIC#2. As can be seen in Figure

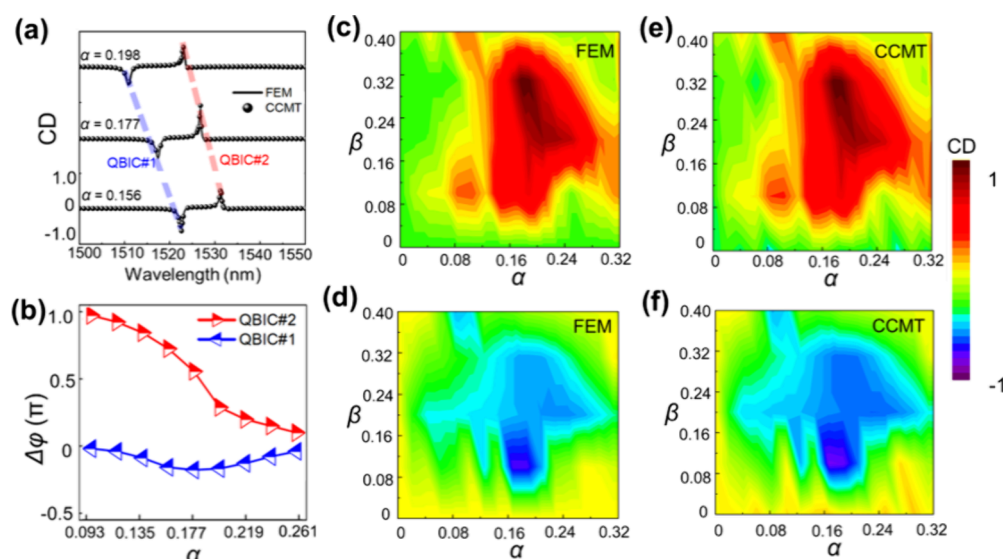


Figure 4. Chiroptical responses of the USM evaluated by using the CCMT. (a) CD responses as functions of α with $\beta = 0.31$ by using FEM and CCMT. (b) Phase difference as a function of α with $\beta = 0.31$ at the resonance wavelengths of QBIC#1 and QBIC#2. (c) CD peak of QBIC#2 and (d) CD dip of QBIC#1 as functions of α and β calculated by using FEM. (e) CD peak of QBIC#2 and (f) CD dip of QBIC#1 as functions of α and β evaluated by using CCMT.

3h, the electric field of the RCP light exhibits significant enhancement compared to its LCP counterpart at the resonance location of QBIC#1. Furthermore, the displacement current vector forms a closed loop at the center of the unit cell of the USM, indicating the characteristic signature of the MD mode. Similarly, the dominant mode is evidenced as the MD mode under the LCP excitation at QBIC#2, through its stronger near-field enhancement and characteristic displacement current loop. Although the MD mode dominates under both LCP and RCP excitations, the rotation directions of the displacement current vector at QBIC#1 and QBIC#2 for both LCP and RCP lights are reversed, resulting in opposite CD signs. These spinning-selective MD modes therefore account for the high-Q intrinsic CD associated with the two QBICs in the USM. Note the dual-band CDs with opposite sign can be maintained even if the structural parameters are significantly altered (see Supporting Information Section 7 and Figure S6).

To provide a general and deeper understanding of the dual-band high-Q CD with opposite sign, we develop a chiral coupled-mode theory (CCMT), previously validated in chiral systems,^{20,43} to analyze the coupling between the USM and incident circularly polarized light. According to the CCMT, the intrinsic CD of the USM can be expressed as (see Supporting Information Section 8)

$$CD = R_{LL} - R_{RR} = \sum_{n=1}^2 \frac{2\gamma_n |\kappa_{x,n}| |\kappa_{y,n}| \sin \Delta\varphi_n}{(\omega - \omega_n)^2 + \gamma_n^2} \quad (3)$$

where n denotes the n th QBIC; ω_n and γ_n are the resonance frequency and decay rate of the n th QBIC, respectively; $\kappa_{x,n}$ and $\kappa_{y,n}$ are the complex coupling parameters for the x and y polarized components of the incident wave, respectively; $\kappa_{x,n} = |\kappa_{x,n}| e^{i\varphi_{x,n}}$, $\kappa_{y,n} = |\kappa_{y,n}| e^{i\varphi_{y,n}}$, and $|\kappa_{x,n}|^2 + |\kappa_{y,n}|^2 = \gamma_n$ where $\varphi_{x,n}$ and $\varphi_{y,n}$ denote the coupling phase shifts,⁴⁴ and the coupling phase difference of reflection for the x and y polarization light is defined as $\Delta\varphi_n = \varphi_{y,n} - \varphi_{x,n}$.

Figure 4a shows the finite element method (FEM) simulation and CCMT results of the CD responses as functions of α with $\beta = 0.31$. As can be seen in Figure 4a,

although the CD responses of the USM are different for different α , the CCMT results are in good agreement with those of FEM. Figure 4b reveals that varying α substantially alters the phase differences of both QBIC#1 and QBIC#2, yet confines them to distinct ranges. The coupling phase difference $\Delta\varphi_2$ of QBIC#2 is confined within $(0, \pi)$, leading to a positive CD, whereas the phase difference $\Delta\varphi_1$ of QBIC#1 is confined to $(-\pi, 0)$, resulting in a negative CD. Note that highly efficient CD can be realized for QBIC#2 around $\alpha = 0.177$ with $\Delta\varphi_2 \approx 0.5\pi$. In contrast, the CD efficiency for QBIC#1 remains comparatively low because its $\Delta\varphi_1$ value deviates significantly from -0.5π . Figure 4c,d illustrates the distributions of the CD peak (QBIC#2) and CD dip (QBIC#1) as functions of α and β , obtained via FEM simulations. The results show that near-perfect CD can be achieved near $(\alpha, \beta) = (0.177, 0.31)$ for QBIC#2, whereas the CD response of QBIC#1 remains comparatively low across the varied parameters. As shown in Figure 4e,f, the excellent agreement between the CCMT calculations and the FEM results validate the CCMT as a reliable framework for characterizing the dual-band high-Q intrinsic CDs with opposite signs in the USM. Note high-efficiency CD of QBIC#2 is achievable even if both α and β are significantly altered, indicating significant tolerance to geometric deviations in asymmetry parameters for the proposed metasurface (see Supporting Information Section 9 and Figure S7). Moreover, the CD performances of the two QBICs show robustness against common fabrication imperfections, including oblique sidewalls and rounded corners (see Supporting Information, Section 10 and Figure S8).

To verify the dual-band high-Q intrinsic CDs with opposite signs of the USM in an experimental setting, we fabricate four types of underetched metasurfaces using EBL and inductively coupled plasma (ICP) etching processes on a 500 nm thick SOI wafer (see Sample Fabrication in the Methods for more details on the fabrication). The SEM images of the representative sample are presented in Figure 5a,b, where $P = 905$ nm, $w = 480$ nm, $\alpha = 0.158$, $\beta = 0.31$, and the samples are $400 \times 400 \mu\text{m}^2$ in size. One can see that the measured parameters of the fabricated metasurface are in good

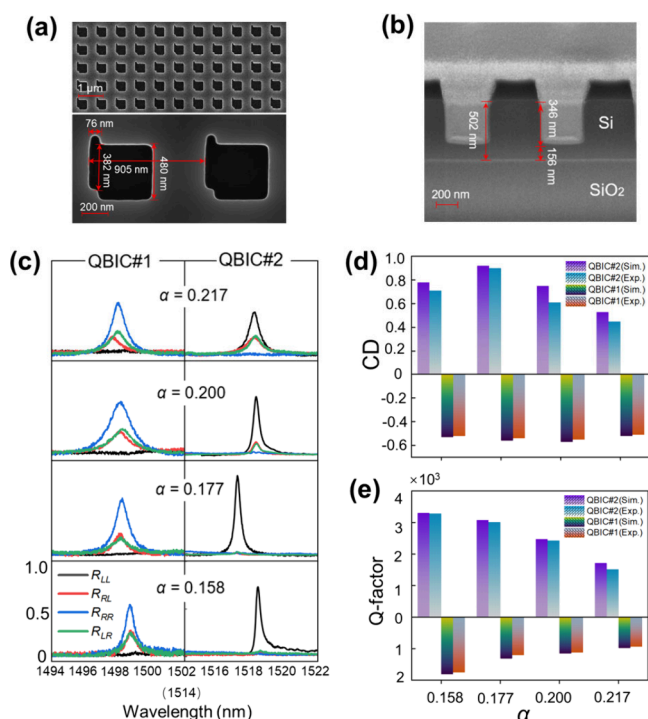


Figure 5. Experimental characterization of high-Q chiroptical responses of the USM. (a) Top-view (top) and zoomed-in (bottom) SEM images of a representative USM with $\alpha = 0.158$ (for samples of other sizes see Supporting Information Section 11 and Figure S9). (b) Side-view SEM image of USM with $\alpha = 0.158$. (c) Experimentally measured reflection components of four fabricated USMs with different α values with β fixed at 0.31. (d) Simulated and experimentally extracted CD values and (e) corresponding Q-factors as functions of α with $\beta = 0.31$.

agreement with the design with slight deviations from smoothed nanohole edges due to fabrication imperfections. Figure 5c illustrates the experimentally measured reflection components (R_{LL} , R_{RL} , R_{RR} , and R_{LR}) for resonances of QBIC#1 and QBIC#2 of the four fabricated USMs, with the measurement setup detailed in the Methods. The spectral responses of QBIC#1 and QBIC#2 are confined in the narrow wavelength ranges with a high-resolution step of ~ 0.02 nm to achieve precise reflection measurements. The resonance locations of QBIC#1 and QBIC#2 are blue-shifted ~ 18 nm and ~ 10 nm, respectively, attributable to structural imperfection in fabrication. As can be seen in Figure 5c, for both QBIC#1 and QBIC#2 across all four samples, the measured cross-polarization components R_{RL} and R_{LR} are nearly identical, whereas copolarization components R_{LL} and R_{RR} differ significantly. This distinct behavior confirms the intrinsic chirality of the USMs. In Figure 5d,e, we compare the simulated and experimentally measured CD values and Q-factors of the QBIC#1 and QBIC#2 resonances as functions of the asymmetry parameter α of the USM. As can be seen in Figure 5d,e, both the QBIC#1 and QBIC#2 resonances exhibit high-Q intrinsic chirality for different α , with their CD values appearing in opposite signs. The measured Q-factors of the QBIC#1 and QBIC#2 resonances can be tuned by varying α , reaching 934 and 1515 at $\alpha = 0.217$, and 1754 and 3282 at $\alpha = 0.158$, respectively. Particularly, near-perfect high-Q intrinsic chirality is experimentally observed for the QBIC#2 resonance at $\alpha = 0.177$, which achieves a high Q-factor of 3012 alongside a strong CD peak of 0.90. In contrast, the QBIC#1 resonance

at $\alpha = 0.200$ exhibits a distinct CD dip of -0.55 with a Q-factor of 1121. The experimental results show excellent agreement with theoretical predictions in both CD values and Q-factor over a wide range of α , confirming that the USM is well-suited for the robust generation of high-Q intrinsic chirality in planar platforms.

To summarize, we have proposed and experimentally validated a Si-based underetched metasurface that exhibits maximal intrinsic chirality mediated by QBIC. The under-etching is introduced as a new degree of freedom to unlock the out-of-plane dimension to achieve intrinsic chirality, thereby enabling the C-points to be precisely tailored to the Γ -point by combining the strategic design of the in-plane perturbation. This design approach enables dual-band high-Q chiral QBICs exhibiting opposite CD signs in the USM. The underlying resonance mechanism, as characterized by vt-RHD and spin-selected MD modes, can be accurately described by CCMT. The opposite CD signs of the two QBICs are attributed to the distinct ranges of the coupling phase difference. Experimental measurements at optical communication wavelengths confirm the pronounced dual-band chiral resonances. Specifically, we observe maximal high-Q intrinsic chirality, characterized by a positive CD of 0.90 ($Q = 3012$) at 1517.03 nm with $\alpha = 0.177$ and a negative CD of -0.55 ($Q = 1121$) at 1497.88 nm with $\alpha = 0.200$. Our findings not only establish underetched metasurfaces as a new paradigm for generating high-Q intrinsic chirality via the QBIC mechanism but also pave the way for advanced chiral light–matter interactions, including chiral lasing, nonlinear chiral optics, and chirality sensing.

NUMERICAL SIMULATION

All of the simulation results in this work were conducted by FEM using COMSOL Multiphysics commercial software. The reflection spectra were simulated under circularly polarized incidence that propagated along the $-z$ -axis direction. Perfectly matched layers were employed in the z direction to mimic a nonreflecting infinite domain. Periodic boundary conditions were applied to the corresponding unit cells along the x - and y -axis directions. The discrete grid was set to the finest resolution to ensure sufficient detail for accurate simulation data.

SAMPLE FABRICATION

The metasurface samples were fabricated on a standard SOI wafer, comprising a 500 nm-thick top silicon layer, a 3.05 μm buried oxide (BOX) layer of SiO_2 , and a 725 μm silicon substrate. The process began with a standard wafer cleaning procedure. Subsequently, an ~ 500 nm-thick layer of ZEP520A resist was spin-coated and prebaked (soft baked) at 180 $^\circ\text{C}$ for 3 min. After pattern exposure using electron-beam lithography, a postexposure bake (PEB) was conducted at 120 $^\circ\text{C}$ for 2 min. The sample subsequently underwent sequential processing including development in the ZED-N50 developer, stopping with ZMD-B fixer. The resulting photoresist pattern served as an etch mask for pattern transfer into the top Si layer via inductively coupled plasma (ICP) etching using a $\text{SF}_6/\text{C}_4\text{F}_8$ gas chemistry. Finally, the residual resist was completely stripped in heated *n*-methyl-2-pyrrolidone (NMP) at 80 $^\circ\text{C}$, followed by a thorough cleaning in acetone and isopropanol. The procedures of the sample fabrication can be seen in Figure S10 in Supporting Information Section 12. Note that the CD performances of the SOI-based metasurfaces are consistent

with the simulations of the metasurfaces on a SiO₂ substrate, as the thickness of the BOX layer of SiO₂ in the SOI wafer is sufficiently large (see Supporting Information Section 13 and Figure S11). It should also be noted that the use of amorphous Si not only causes a shift in the CD peak but also reduces CD efficiency due to absorption (see Supporting Information Section 14 and Figure S12).

■ EXPERIMENTAL SETUPS

The reflection spectra of the fabricated devices were measured using a home-built setup (see Figure S13 in Supporting Information, Section 15). A supercontinuum laser (NKT SuperK) served as the light source. The CPL was generated by passing the laser through a linear polarizer (LP1) and a quarter-wave plate (QWP1). A lens (L1) was used to focus the incident light on the rear focal plane of the objective to avoid the extra incident angle from the objective (MY10X-823, NA = 0.26, 10× magnification, and a working distance of 30.5 mm). The reflected light was collected by the same objective lens and divided by a 50:50 beam splitter. The reflected light was analyzed by another set of a quarter-wave plate (QWP2) and a linear polarizer (LP2) to resolve its copolarized (R_{LL} and R_{RR}) and cross-polarized (R_{RL} and R_{LR}) components. And then the divided reflected light was focused by a lens (L2) and collected by a spectrometer (ANDOR SR750) and camera, separately. The measured intensities for each polarization configuration were normalized to those from a calibrated reference mirror to obtain the reflected components. Measurements were conducted in a darkroom to reduce the impact of background light.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.6c00556>.

Supplementary text and Figures S1–S13 including additional theoretical, simulation, and experimental details to support the main findings: detailed mode characteristics, far-field properties of the overetched metasurfaces, comparison between RHD and OCD, chiroptical responses in transmission, reciprocal excitation behaviors, multipole decomposition analysis, structural parameter dependencies, chiral coupled-mode theory derivations, robustness of the CD efficiency, CD performances under fabrication imperfections, supplementary SEM images, influence of the BOX layer thickness, CD response in amorphous silicon, and schematic diagrams of the fabrication process and measurement device (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Guanhai Li – State Key Laboratory of Infrared Physics, Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai 200083, China; University of Chinese Academy of Science, Beijing 100049, China; Hangzhou Institute for Advanced Study, University of Chinese Academy of Sciences, Xiangshan, Hangzhou 310024, China; Shanghai Research Center for Quantum Sciences, Shanghai 201315, China; orcid.org/0000-0002-2745-055X; Email: ghli0120@mail.sitp.ac.cn

Tian Sang – Department of Optoelectronic Information Science and Engineering, School of Optoelectronic Information and Physical Science, Jiangnan University, Wuxi 214122, China; orcid.org/0000-0002-9847-6001; Email: sangt@jiangnan.edu.cn

Authors

Leyi Zhu – Department of Optoelectronic Information Science and Engineering, School of Optoelectronic Information and Physical Science, Jiangnan University, Wuxi 214122, China

Yukang Zhang – State Key Laboratory of Infrared Physics, Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai 200083, China

Guanhua Yang – Department of Optoelectronic Information Science and Engineering, School of Optoelectronic Information and Physical Science, Jiangnan University, Wuxi 214122, China

Rong Jin – State Key Laboratory of Infrared Physics, Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai 200083, China

Chenyang Wang – Department of Optoelectronic Information Science and Engineering, School of Optoelectronic Information and Physical Science, Jiangnan University, Wuxi 214122, China

Yueke Wang – Department of Optoelectronic Information Science and Engineering, School of Optoelectronic Information and Physical Science, Jiangnan University, Wuxi 214122, China

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.nanolett.6c00556>

Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Mun, J.; Kim, M.; Yang, Y.; Badloe, T.; Ni, J.; Chen, Y.; Qiu, C.-W.; Rho, J. Electromagnetic Chirality: from Fundamentals to Nontraditional Chiroptical Phenomena. *Light: Sci. Appl.* **2020**, *9* (1), 139.
- (2) Lininger, A.; Palermo, G.; Guglielmelli, A.; Nicoletta, G.; Goel, M.; Hinczewski, M.; Strangi, G. Chirality in Light-Matter Interaction. *Adv. Mater.* **2023**, *35* (34), 2107325.
- (3) Kim, Y.; Kim, H.; Yang, Y.; Badloe, T.; Jeon, N.; Rho, J. Three-Dimensional Artificial Chirality towards Low-cost and Ultra-sensitive Enantioselective Sensing. *Nanoscale* **2022**, *14* (10), 3720–3730.
- (4) Sharma, S.; Karabchevsky, A.; Varshney, S. K. Accidental BICs and Chiroptical Response of Hollow Channels in Silicon Nanodisk Resonator-Based Metasurfaces for Chiral Sensing. *ACS Appl. Opt. Mater.* **2025**, *3* (2), 364–374.
- (5) Zhang, L.; Jia, F.; Xu, L.; Rahmani, M.; Qin, T.; Mei, T.; Zhang, W. Intrinsically Chiral Metamaterials with Giant Optical Chirality via One-Step Lithography. *Adv. Funct. Mater.* **2025**, No. e24806.
- (6) Jia, T.; Jeon, Y.; Feng, L.; Kim, H.; Li, B.; Rui, G.; Rho, J. Superchirality Induced Ultrasensitive Chiral Detection in High-Q Optical Cavities. *Opto-Electron. Adv.* **2025**, *8* (10), 250079.
- (7) Hentschel, M.; Schäferling, M.; Duan, X.; Giessen, H.; Liu, N. Chiral Plasmonics. *Sci. Adv.* **2017**, *3* (5), No. e1602735.

- (8) Li, J.; Yu, P.; Zhang, S.; Liu, N. Electrically-Controlled Digital Metasurface Device for Projection Displays. *Nat. Commun.* **2020**, *11* (1), 3574.
- (9) Shi, T.; Deng, Z.-L.; Geng, G.; Overvig, A.; Li, J.; Qiu, C.-W.; Zeng, X.; Zeng, Y.; Hu, G.; Alù, A.; Kivshar, Y. S.; Li, X. Planar Chiral Metasurfaces with Maximal and Tunable Chiroptical Response Driven by Bound States in the Continuum. *Nat. Commun.* **2022**, *13* (1), 4111.
- (10) Oshikiri, T.; Sun, Q.; Yamada, H.; Zu, S.; Sasaki, K.; Misawa, H. Extrinsic Chirality by Interference between Two Plasmonic Modes on an Achiral Rectangular Nanostructure. *ACS Nano* **2021**, *15* (10), 16802–16810.
- (11) Petronijevic, E.; Cesca, T.; Scian, C.; Mattei, G.; Voti, R. L.; Sibilia, C.; Belardini, A. Extrinsic Chirality Tailors Stokes Parameters in Simple Asymmetric Metasurfaces. *Nanoscale* **2024**, *16* (35), 16477.
- (12) Wu, C.; Arju, N.; Kelp, G.; Fan, J.; Dominguez, J.; Gonzales, E.; Tutuc, E.; Brener, I.; Shvets, G. Spectrally Selective Chiral Silicon Metasurfaces Based on Infrared Fano resonances. *Nat. Commun.* **2014**, *5* (1), 3892.
- (13) Gansel, J. K.; Thiel, M.; Rill, M. S.; Decker, M.; Bade, K.; Saile, V.; von Freymann, G.; Linden, S.; Wegener, M. Gold Helix Photonic Metamaterial as Broadband Circular Polarizer. *Science* **2009**, *325* (5947), 1513–1515.
- (14) Lin, Z.; Mikhael, C.; Dai, C.; Cho, J.-H. Self-Assembly for Creating Vertically-Aligned Graphene Micro Helices with Monolayer Graphene as Chiral Metamaterials. *Adv. Mater.* **2024**, *36* (27), 2401451.
- (15) Choi, W.; Moestopo, W. P.; Gu, S.; Lee, T.; Yoo, J.-H.; Xia, X.; Armstrong, M. R. Helical Photonic Metamaterials for Encrypted Chiral Holograms. *Adv. Sci.* **2025**, *12*, No. e07931.
- (16) Dietrich, K.; Lehr, D.; Helgert, C.; Tünnermann, A.; Kley, E. B. Circular Dichroism from Chiral Nanomaterial Fabricated by On-Edge Lithography. *Adv. Mater.* **2012**, *24* (44), OP321–OP325.
- (17) Chen, Q.; Wu, Y.; She, Y.; Zhao, Y.; Yang, J.; Chen, L.; Liu, P.; Wu, M.; Zeng, C.; Qi, Z.; Hu, C.; Liu, H.; Xiong, Y.; Tian, Y.; Chen, Y.; Cai, H.; Pan, N.; Wang, X. Continuously Geometrical Tuning to Boost Circular Dichroism in 3D Chiral Metamaterials. *Adv. Opt. Mater.* **2024**, *12* (21), 2400593.
- (18) Gorkunov, M. V.; Antonov, A. A.; Kivshar, Y. S. Metasurfaces with Maximum Chirality Empowered by Bound States in the Continuum. *Phys. Rev. Lett.* **2020**, *125* (9), 093903.
- (19) Overvig, A.; Yu, N.; Alù, A. Chiral Quasi-Bound States in the Continuum. *Phys. Rev. Lett.* **2021**, *126* (7), 073001.
- (20) Gorkunov, M. V.; Antonov, A. A.; Tuz, V. R.; Kupriianov, A. S.; Kivshar, Y. S. Bound States in the Continuum Underpin Near-Lossless Maximum Chirality in Dielectric Metasurfaces. *Adv. Opt. Mater.* **2021**, *9* (19), 2100797.
- (21) Tang, Y.; Liang, Y.; Yao, J.; Chen, M. K.; Lin, S.; Wang, Z.; Zhang, J.; Huang, X.; Yu, C.; Tsai, D. Chiral Bound States in the Continuum in Plasmonic Metasurfaces. *Laser Photonics Rev.* **2023**, *17* (4), 2200597.
- (22) Zhang, X.; Liu, Y.; Han, J.; Kivshar, Y.; Song, Q. Chiral Emission from Resonant Metasurfaces. *Science* **2022**, *377* (6611), 1215–1218.
- (23) Chen, Y.; Deng, H.; Sha, X.; Chen, W.; Wang, R.; Chen, Y.-H.; Wu, D.; Chu, J.; Kivshar, Y. S.; Xiao, S.; Qiu, C.-W. Observation of Intrinsic Chiral Bound States in the Continuum. *Nature* **2023**, *613* (7944), 474–478.
- (24) Sha, X.; Du, K.; Zeng, Y.; Lai, F.; Yin, J.; Zhang, H.; Song, B.; Han, J.; Xiao, S.; Kivshar, Y.; Song, Q. Chirality Tuning and Reversing with Resonant Phase-Change Metasurfaces. *Sci. Adv.* **2024**, *10* (21), No. eadn901.
- (25) Lim, Y.; Seo, I. C.; An, S.-C.; Kim, Y.; Park, C.; Woo, B. H.; Kim, S.; Park, H.-R.; Jun, Y. C. Maximally Chiral Emission via Chiral Quasibound States in the Continuum. *Laser Photonics Rev.* **2023**, *17* (2), 2200611.
- (26) Kühner, L.; Wendisch, F. J.; Antonov, A. A.; Bürger, J.; Hüttenhofer, L.; de S. Menezes, L.; Maier, S. A.; Gorkunov, M. V.; Kivshar, Y.; Tittl, A. Unlocking the Out-of-Plane Dimension for Photonic Bound States in the Continuum to Achieve Maximum Optical Chirality. *Light Sci. Appl.* **2023**, *12*, 250.
- (27) Koshelev, K.; Lepeshov, S.; Liu, M.; Bogdanov, A.; Kivshar, Y. Asymmetric Metasurfaces with High-Q Resonances Governed by Bound States in the Continuum. *Phys. Rev. Lett.* **2018**, *121* (19), 193903.
- (28) Zhou, C.; Huang, L.; Jin, R.; Xu, L.; Li, G.; Rahmani, M.; Chen, X.; Lu, W.; Miroshnichenko, A. E. Bound States in the Continuum in Asymmetric Dielectric Metasurfaces. *Laser Photonics Rev.* **2023**, *17* (3), 2200564.
- (29) Sang, T.; Mi, Q.; Yang, C.; Zhang, X.; Wang, Y.; Ren, Y.; Xu, T. Achieving Asymmetry Parameter-Insensitive Resonant Modes through Relative Shift-Induced Quasi-Bound States in the Continuum. *Nanophotonics* **2024**, *13* (8), 1369–1377.
- (30) He, H.; Lai, F.; Zhang, Y.; Zhang, X.; Tian, X.; Li, Y.; Wang, Y.; Xiao, S.; Huang, L. Spectro-Polarimetric Detection Enabled by Multidimensional Metasurface with Quasi-Bound States in the Continuum. *Opto-Electron. Adv.* **2025**, *8* (10), 250015.
- (31) Jeon, D.; Rho, J. Quasi-Trapped Guided Mode in a Metasurface Waveguide for Independent Control of Multiple Nonlocal Modes. *ACS Photonics* **2024**, *11* (2), 703–713.
- (32) Wang, B.; Liu, W. Z.; Zhao, M. X.; Wang, J. J.; Zhang, Y. W.; Chen, A.; Guan, F.; Liu, X. H.; Shi, L.; Zi, J. Generating Optical Vortex Beams by Momentum-Space Polarization Vortices Centred at Bound States in the Continuum. *Nat. Photonics* **2020**, *14* (10), 623–628.
- (33) Jing, Z.; Li, S.; Ouyang, S.; Lu, J.; Wang, Y.; Huang, L.; Li, L.; Sang, T. Observation of the Generalized Kerker Effect Mediated by Quasi-Bound States in the Continuum. *Nano Lett.* **2025**, *25* (1), 522–528.
- (34) Qin, H.; Su, Z.; Liu, M.; Zeng, Y.; Tang, M.-C.; Li, M.; Shi, Y.; Huang, W.; Qiu, C.-W.; Song, Q. Arbitrarily Polarized Bound States in the Continuum with Twisted Photonic Crystal Slabs. *Light: Sci. Appl.* **2023**, *12* (1), 66.
- (35) Bliokh, K. Y.; Kivshar, Y. S.; Nori, F. Magnetolectric Effects in Local Light-Matter Interactions. *Phys. Rev. Lett.* **2014**, *113* (3), 033601.
- (36) Nieto-Vesperinas, M.; Xu, X. Reactive Helicity and Reactive Power in Nanoscale Optics: Evanescent Waves. Kerker Conditions. Optical Theorems and Reactive Dichroism. *Phys. Rev. Res.* **2021**, *3* (4), 043080.
- (37) Chen, W.; Wang, Z.; Gorkunov, M. V.; Qin, J.; Wang, R.; Wang, C.; Wu, D.; Chu, J.; Wang, X.; Kivshar, Y.; Chen, Y. Uncovering Maximum Chirality in Resonant Nanostructures. *Nano Lett.* **2024**, *24* (31), 9643–9649.
- (38) Lu, J.; Sang, T.; Pian, C.; Ouyang, S.; Jing, Z. Tailoring Intrinsic Chiroptical Responses via Twisted Bilayer α -MoO₃ Separated by a VO₂ Film. *APL Photonics* **2024**, *9* (4), 046112.
- (39) Khaliq, H. S.; Kim, M.-S.; Kim, H.-R. Ultrahigh-Q Spin-selective Dual Quasi-BIC Resonances in Ultraviolet Dielectric Metasurfaces. *Mater. Des.* **2025**, *256*, 114343.
- (40) Li, S.; Sang, T.; Yang, C.; Lu, J.; Wang, Y. Phase-Change Metasurfaces for Dynamic Control of Chiral Quasi-Bound States in the Continuum. *Opt. Lett.* **2023**, *48* (24), 6488–6491.
- (41) Zhu, A. Y.; Chen, W. T.; Zaidi, A.; Huang, Y.-W.; Khorasaninejad, M.; Sanjeev, V.; Qiu, C.-W.; Capasso, F. Giant Intrinsic Chiro-Optical Activity in Planar Dielectric Nanostructures. *Light: Sci. Appl.* **2018**, *7* (2), 17158–17158.
- (42) Hassanfirooz, A.; Lu, Y. C.; Wu, P. C. Hybrid Anapole Induced Chirality in Metasurfaces. *Adv. Mater.* **2024**, *36* (46), 2410568.
- (43) Cao, Z.; Yiu, L.; Zhang, Z.; Chan, C. T.; Ong, H. Understanding the Role of Surface Plasmon Polaritons in Two-Dimensional Achiral Nanohole Arrays for Polarization Conversion. *Phys. Rev. B* **2017**, *95* (15), 155415.
- (44) Ji, C.-Y.; Chen, S.; Han, Y.; Liu, X.; Liu, J.; Li, J.; Yao, Y. Artificial Propeller Chirality and Counterintuitive Reversal of Circular Dichroism in Twisted Meta-Molecules. *Nano Lett.* **2021**, *21* (16), 6828–6834.